

Karate Championship App

Developer environment setup guide · Windows 11 · winget-based installation

Prerequisites: Windows 11 with winget available (built-in since Windows 10 1709+). Run all winget commands in a terminal with administrator privileges. Open a new terminal window after each installation to pick up PATH changes.

STEP 1 — JAVA RUNTIME

JDK 21 (Adoptium Temurin)

Version: 21.x LTS · Free, GPL licensed · Required by Spring Boot 3.x

```
winget install EclipseAdoptium.Temurin.21.JDK
```

After install: set JAVA_HOME system variable pointing to the JDK folder. Add %JAVA_HOME%\bin to system PATH. Verify: java -version

Why Adoptium and not Oracle JDK? Adoptium Temurin is a certified OpenJDK build with identical bytecode compatibility and no licensing restrictions for personal or commercial use. Oracle JDK requires a paid licence in production commercial environments.

STEP 2 — BUILD TOOL

Apache Maven 3.9.x

Version: 3.9.x · Free · No winget package — manual install required

Download binary zip: <https://maven.apache.org/download.cgi>

Extract the zip to a folder of your choice. Set MAVEN_HOME system variable to the extracted folder. Add %MAVEN_HOME%\bin to system PATH. Verify: mvn -version

Maven stores downloaded dependencies in C:\Users\...\m2\repository by default. To redirect this to another location, edit conf\settings.xml inside the Maven folder and set: YOUR_PREFERRED_PATH\m2\repository

STEP 3 — CONTAINERISATION

Docker Desktop

Version: latest · Free for personal use · WSL2 backend recommended

```
winget install Docker.DockerDesktop
```

During install: select 'Use WSL 2 instead of Hyper-V'. Hardware virtualisation must be enabled in BIOS (Intel VT-x or AMD SVM/AMD-V). After install: Settings → Resources → Advanced → set disk image location to your preferred drive.

This project runs MongoDB inside a Docker container — no local MongoDB service needed on the OS. MongoDB Compass (GUI client, step 5) connects to the container on localhost:27017 identically to a local install. Docker also sets you up for the deployment milestone: the finished app will run as composed containers (backend + frontend + database) via a single docker-compose.yml.

STEP 4 — IDE

IntelliJ IDEA

Version: latest (2025.3+ unified release) · Core features free, Ultimate via subscription

```
winget install JetBrains.IntelliJIDEA.Community
```

As of December 2025, Community and Ultimate are one unified product — one installer, one download. Core features (Java, Spring Boot, Maven, JUnit) are fully free. Ultimate features unlock with a 30-day trial or paid subscription. Students with a university email can get Ultimate free at jetbrains.com/community/education

STEP 5 — API TESTING & DATABASE GUI

Postman

Version: latest · Free tier available · REST API testing

```
winget install Postman.Postman
```

MongoDB Compass

Version: latest · Free · Connects to Docker-hosted MongoDB on localhost:27017

```
winget install MongoDB.Compass.Full
```

STEP 6 — AI CODING ASSISTANT (OPTIONAL)

This step is entirely optional. Any AI coding assistant of your choice can be used alongside this project. Node.js is only required as a runtime for the Claude Code CLI — if you choose a different tool, skip this step entirely.

Node.js

Version: LTS · Only needed as a runtime for Claude Code

```
winget install OpenJS.NodeJS.LTS
```

After install: optionally redirect global npm packages to a non-system drive: npm config set prefix "YOUR_PATH\npm-global" — then add that path to system PATH.

OPTIONAL

Claude Code CLI

AI coding assistant by Anthropic · Requires Node.js

```
npm install -g @anthropic-ai/claude-code
```

Verify: `claude --version` · Requires an Anthropic account. Other AI assistants (GitHub Copilot, Cursor, etc.) can be used instead.

OPTIONAL

POST-INSTALL CONFIGURATION SUMMARY

JAVA_HOME	Path to JDK 21 installation folder
MAVEN_HOME	Path to extracted Maven folder
PATH additions	%JAVA_HOME%\bin and %MAVEN_HOME%\bin (+ npm global if using Claude Code)
Maven local repo	Optional – redirect via conf\settings.xml in Maven folder
Docker data	Optional – redirect in Docker Desktop: Settings → Resources → Advanced
MongoDB connection	localhost:27017 – connect via Compass or Postman once container is running

PROJECT STACK REFERENCE

Language	Java 21
Framework	Spring Boot 3.2.3
Build tool	Maven 3.9.x
Database	MongoDB (Docker container)
API layer	REST – tested via Postman
Testing	JUnit (spring-boot-starter-test)

Karate Championship Management App · github.com/AthleticAvi/Karate-Championship-Management-App